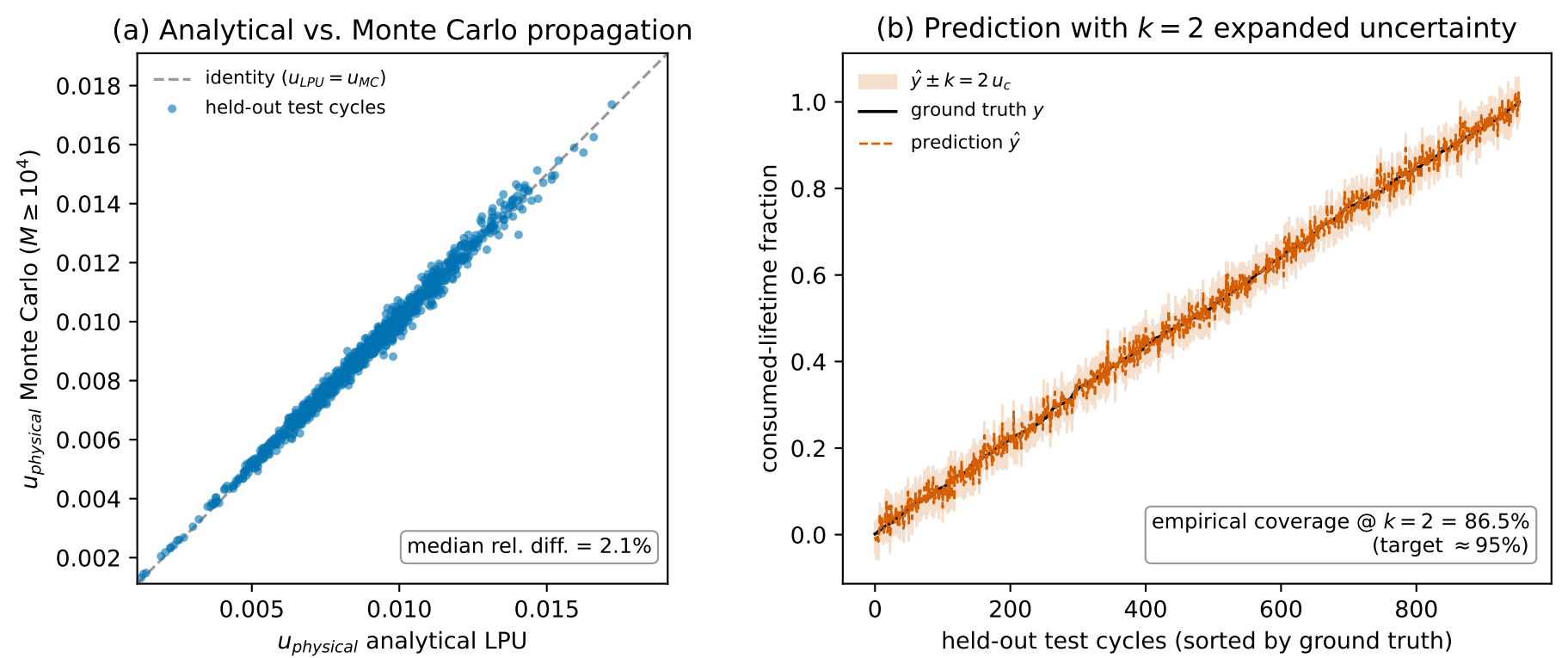

(a) Analytical vs. Monte Carlo propagation
identity ($u_{LPU} = u_{MC}$)
held-out test cycles
median rel. diff. = 2.1%
$u_{physical}$ Monte Carlo ($M \geq 10^4$)
$u_{physical}$ analytical LPU
0.002
0.004
0.006
0.008
0.010
0.012
0.014
0.016
0.018
0.005
0.010
0.015
(b) Prediction with $k = 2$ expanded uncertainty
$\hat{y} \pm k = 2\,u_c$
ground truth y
prediction $\hat{y}$
empirical coverage @ $k = 2$ = 86.5%
(target ≈ 95%)
consumed-lifetime fraction
held-out test cycles (sorted by ground truth)
0.0
0.2
0.4
0.6
0.8
1.0
0
200
400
600
800